

# GOSFORD HIGH SCHOOL



**2009**

**Trial HSC**

## MATHEMATICS EXTENSION I

*Time Allowed: 2 Hours + 5 minutes reading time*

### **General Instructions:**

- Reading Time – 5 minutes.
- Working time – 2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.
- Each question should be started in a separate writing booklet.

### **TOTAL MARKS – 84**

- Attempt Questions 1 – 7
- All questions are of equal value.

# Mathematics Extension 1 Trial Higher School Certificate - 2009

Marks

## Question 1 (12 Marks)

- a) Solve  $\frac{2x-3}{x-2} \geq 1$  2
- b) The point P divides the segment AB externally in the ratio 3:2.  
If A is the point (1, 4), and B is the point (-1, 8), find the coordinates of P. 2
- c) State the domain and range of  $y = \cos^{-1} \frac{x}{3}$  2
- d) Evaluate  $\int_0^{\frac{\pi}{4}} \cos^3 \theta \sin \theta d\theta$  2
- e) The polynomial  $x^4 + 2x$  is divided by  $x+2$ . Calculate the remainder. 2
- f) The acute angle between  $y = 3x + 5$  and  $y = mx + 4$  is  $45^\circ$ . 2  
Find 2 possible values of m.

**Question 2 (12 Marks) Begin a new booklet.**

**Marks**

a) Differentiate  $x \tan^{-1} 3x$  with respect to  $x$ . **2**

b) Find  $\int \sin^2 3x \, dx$  **3**

c) Evaluate  $\int_0^3 \frac{x}{\sqrt{x+1}} dx$  using the substitution  $x = u^2 - 1$  **3**

d) The polynomial  $P(x) = x^5 + ax^3 + bx$  (where  $a$  and  $b$  are numerical constants), leaves a remainder of 5 when divided by  $x - 2$ .

(i) Show that  $P(x)$  is odd. **1**

(ii) Find the remainder when  $P(x)$  is divided by  $x + 2$ . **1**

e) The equation  $x^3 + 2x^2 + 3x + 6 = 0$  has roots  $\alpha, \beta$  and  $\gamma$ . **2**

Find the value of  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ .

**Question 3 (12 Marks) Begin a new booklet.**

**Marks**

- a) (i) Find the derivative of  $\sin^{-1} x + \cos^{-1} x$  **1**
- (ii) Hence, find the value of  $\sin^{-1} x + \cos^{-1} x$  for all  $x$ . **1**
- b) Taking  $x = 0.5$  as a first approximation to the root of  $x + \ln x = 0$ , **2**  
use Newton's method to find a second approximation.
- c) (i) Sketch the graph of  $y = |1 - 2x|$  **1**
- (ii) Hence, or otherwise, solve  $|1 - 2x| \leq x$  **3**
- d) Prove by mathematical induction that **4**
- $$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n$$
- for all integers  $n \geq 1$

**Question 4 (12 Marks) Begin a new booklet.**

**Marks**

- a) (i) Express  $\sqrt{3}\cos 2t - \sin 2t$  in the form  $A\cos(2t + \alpha)$ , **2**  
with  $A > 0$  and  $\alpha$  acute.

- (ii) Find, in exact form, the general solutions to **2**

$$\sqrt{3}\cos 2t - \sin 2t = 1$$

- b) Two points  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  lie on the parabola  $x^2 = 4ay$  where  $a > 0$ . The chord PQ passes through the focus.

- (i) Show that  $pq = -1$ . **1**

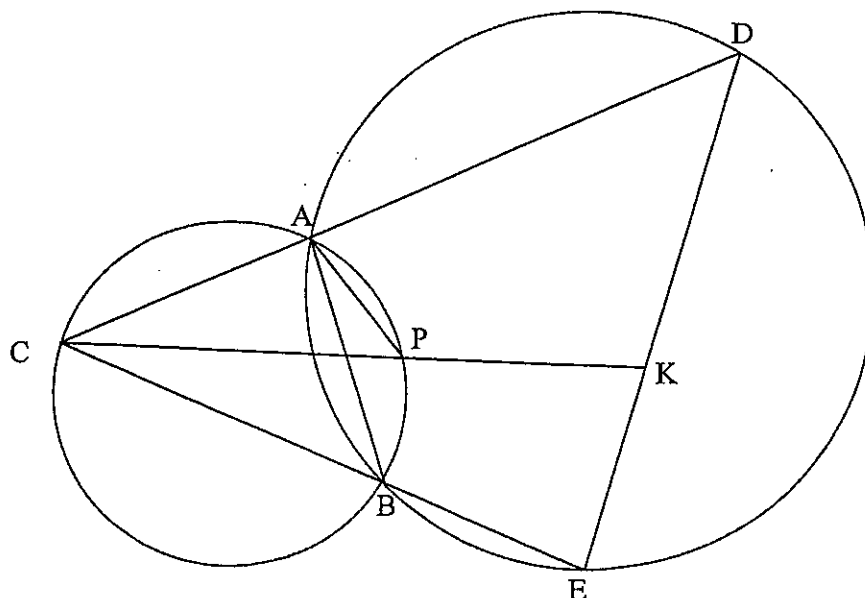
- (ii) Show that the point of intersection T of the tangents to the parabola at P and Q lies on the line  $y = -a$ . **2**

- (iii) Show that the chord PQ has length  $a(p + \frac{1}{p})^2$ . **2**  
(Hint: Use the focus-directrix definition of the parabola)

- c) In the diagram, two circles intersect at A and B. CAD, CBE, CPK and DKE are straight lines. **1**

- i) Show why  $\angle APC = \angle ABC$  **1**

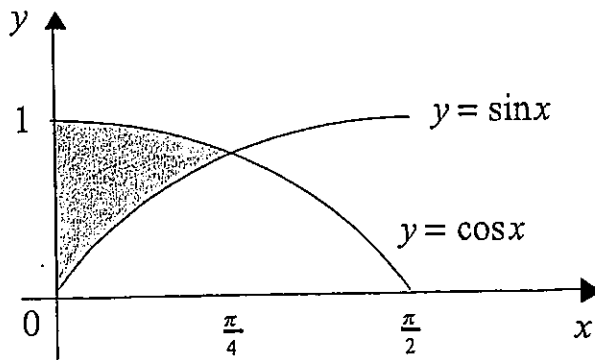
- ii) Hence, or otherwise, show that ADKP is a cyclic quadrilateral. **2**



**Question 5 (12 Marks) Begin a new booklet.**

**Marks**

a)



The region bounded by the curves  $y = \cos x$  and  $y = \sin x$  between  $x = 0$  and  $x = \frac{\pi}{4}$  is rotated through one complete revolution about the  $x$ -axis. Find the volume of the solid. 2

b) A particle, whose displacement is  $x$ , moves in simple harmonic motion such that

$$\frac{d^2x}{dt^2} = -16x. \quad \text{At time } t = 0, \quad x = 1 \text{ and } \frac{dx}{dt} = 4.$$

(i) Show that, for all positions of the particle, 2

$$\left| \frac{dx}{dt} \right| = 4\sqrt{2-x^2}$$

(ii) What is the particle's greatest displacement? 1

(iii) Find  $x$  as a function of  $t$ . You may assume the general form for  $x$ . 2

c) A rectangular vessel is divided into two equal compartments by a vertical porous membrane. Liquid in one compartment, initially at a depth of 20 cm, passes into the other compartment, initially empty, at a rate proportional to the difference in levels.

(i) If the depth of liquid in one of the vessels at any time  $t$  minutes is  $x$  cm, show that

$$\frac{dx}{dt} = k(20 - 2x) \quad \text{1}$$

(ii) Show that  $x = 10(1 - e^{-2kt})$  is a solution to this equation 1

(iii) If the level in the second compartment rises 2 cm in the first 5 minutes, after what time will the difference in levels be 2 cm? 3

**Question 6 (12 Marks) Begin a new booklet.****Marks**

- a) A spherical bubble is expanding so that its volume is increasing at the constant rate of  $10 \text{ mm}^3$  per second. What is the rate of increase of the radius when the surface area is  $500 \text{ mm}^2$ ? **2**

- b) A particle moves in a straight line. Initially it is 2 m to the right of a fixed point O, and velocity is  $v \text{ m/s}$  where

$$v = \frac{32}{x} - \frac{x}{2}$$

- (i) Prove that  $\frac{d^2x}{dt^2} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$  **2**

- (ii) Find an expression for acceleration in terms of  $x$ . **2**

- (iii) Show that  $t = \int \frac{2x}{64 - x^2} dx$  **3**

and hence show  $x^2 = 64 - 60e^{-t}$

- (iv) Sketch the graph of  $x^2$  against  $t$  and describe the limiting behaviour of the particle. **3**

**Question 7 (12 Marks) Begin a new booklet.**

**Marks**

a) Sketch the graph of  $y = \sec x$  for  $-\pi \leq x \leq 2\pi$

**2**

b) The inverse secant  $y = \sec^{-1} x$  could be defined as the function

$$x = \sec y \text{ with } 0 \leq y \leq \pi \text{ and } y \neq \frac{\pi}{2}$$

i) Find  $\sec^{-1} \sqrt{2}$

**1**

(ii) Sketch the graph of  $y = \sec^{-1} x$

**2**

(iii) For  $y = \sec^{-1} x$  find  $\frac{dx}{dy}$  and hence show that  $\frac{dy}{dx} = \left| \frac{1}{x\sqrt{x^2 - 1}} \right|$

**3**

Why is the absolute value sign appropriate?

**1**

(iv) Hence, differentiate  $y = \sec^{-1} \frac{x}{a}$  and simplify your answer.

**1**

(v) Hence, or otherwise, find  $\int \frac{dx}{x\sqrt{25x^2 - 9}}$  for positive  $x$ .

**2**

**End of Examination**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$

# 2009 Exd 1 Trial

## Question 1

a)  $\frac{2x-3}{x-2} \geq 1$

For  $x > 2$  Solve

$$2x-3 \geq x-2$$

$$x \geq 1$$

$$\therefore \text{All } x > 2$$

For  $x < 2$  Solve

$$2x-3 \leq x-2$$

$$x \leq 1$$

$$\therefore x \leq 1$$

$$\therefore \underline{x \leq 1 \text{ or } x > 2}$$

b)  $A(x_1, y_1) = (1, 4)$   $B(x_2, y_2) = (-1, 8)$   $\frac{h}{c} = \frac{3}{-2}$

$$x = \frac{hx_2 + lx_1}{h+l}$$

$$y = \frac{hy_2 + ly_1}{h+l}$$

$$x = \frac{3x_2 + 2x_1}{3+2}$$

$$y = \frac{3y_2 + 2y_1}{3+2}$$

$$P \text{ is } (-5, 16)$$

c)  $-1 \leq \frac{x}{3} \leq 1$

$$\therefore D: -3 \leq x \leq 3$$

$$R: 0 \leq y \leq \pi$$

d)  $\left[ -\frac{\cos^4 \theta}{4} \right]_0^{\frac{\pi}{4}}$   
 $= -\frac{1}{4} \left( \left( \frac{1}{\sqrt{2}} \right)^4 - 1 \right)$   
 $= -\frac{1}{4} \left( \frac{1}{2} - 1 \right)$   
 $= \frac{3}{16}$

e)  $P(x) = x^4 + 2x$   
 $P(-2) = (-2)^4 + 2(-2)$   
 $= 16 - 4$   
 $= 12$

f)  $m_1 = 3$ ,  $m_2 = m$   
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$1 = \left| \frac{3 - m}{1 + 3m} \right|$$

$$|1 + 3m| = |3 - m|$$

$$\therefore 1 + 3m = 3 - m \quad \text{or} \quad 1 + 3m = m - 3$$

$$4m = 2$$

$$2m = -4$$

$$m = \frac{1}{2}$$

$$m = -2$$

## Question 2

a)  $y' = x \cdot \frac{3}{1+9x^2} + \tan^{-1} 3x$

$$y' = \frac{3x}{1+9x^2} + \tan^{-1} 3x$$

b)

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$2\sin^2 \theta = 1 - \cos 2\theta$$

$$\sin^2 \theta = \frac{1}{2} (1 - \cos 2\theta)$$

$$\therefore \int \sin^2 3x \, dx = \frac{1}{2} \int (1 - \cos 6x) \, dx$$

$$= \frac{1}{2} \left( x - \frac{\sin 6x}{6} \right)$$

$$= \frac{x}{2} - \frac{\sin 6x}{12} + C$$

c)  $\int_0^3 \frac{x}{\sqrt{x+1}} \, dx$

$$= \int_1^2 \frac{(u^2-1) \cdot 2u \, du}{u}$$

$$= 2 \int_1^2 (u^2 - 1) \, du$$

$$= 2 \left[ \frac{u^3}{3} - u \right]_1^2$$

$$= 2 \left[ \left( \frac{8}{3} - 2 \right) - \left( \frac{1}{3} - 1 \right) \right]$$

$$= 2 \left( \frac{8}{3} - 2 - \frac{1}{3} + 1 \right)$$

$$= \frac{8}{3}$$

$$x = u^2 - 1 \text{ or } u = \sqrt{x+1}$$

$$dx = 2u \, du$$

$$x=0, u = \sqrt{1} = 1$$

$$x=3, u = \sqrt{4} = 2$$

(positive  $\sqrt{\phantom{x}}$  as  $\sqrt{x+1}$  is +ve)

Q2 (cont)

d)  $P(x) = x^5 + ax^3 + bx$

$$P(-x) = (-x)^5 + a(-x)^3 + b(-x)$$

$$= -x^5 - ax^3 - bx$$

$$= -P(x)$$

$\therefore P(x)$  is odd.

i)  $P(-2) = -P(2)$   
 $= -5$

e)  $x^3 + 2x^2 + 3x + 6 = 0$

$$\alpha + \beta + \gamma = -2$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 3$$

$$\alpha\beta\gamma = -6$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{3}{-6}$$

$$= -\frac{1}{2}$$

$$3) a) i) \quad y' = \frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}} = 0$$

$$ii) \quad \therefore \sin^{-1} x + \cos^{-1} x = \text{constant}$$

$$\text{Sub } x = 0$$

$$\sin^{-1} 0 + \cos^{-1} 0 = c$$

$$0 + \frac{\pi}{2} = c$$

$$\therefore \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$b) \quad f(x) = x + \ln x$$

$$f'(x) = 1 + \frac{1}{x}$$

$$f(0.5) = -0.193$$

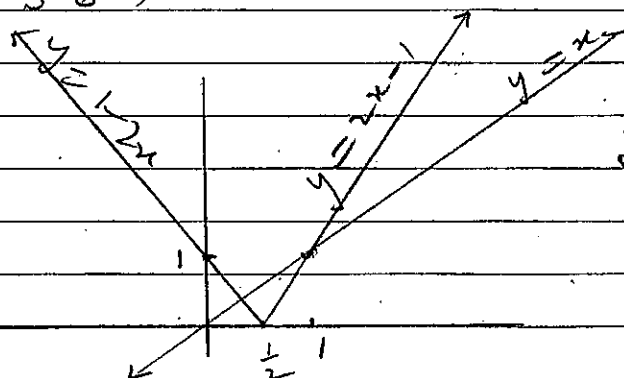
$$f'(0.5) = 3$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 0.5 - \frac{-0.193}{3}$$

$$x_2 = 0.564$$

c) i)



$$\text{Solve } 2x-1=x \\ x=1$$

$$1-2x=x \\ x=\frac{1}{3}$$

$$\text{From graph } \frac{1}{3} \leq x \leq 1$$

$$d) \text{ Prove true for } n=1: \quad \text{LHS} = 1 \times 2^0 = 1 \quad \text{RHS} = 1 + 0 = 1$$

Assume true for  $n=k$ :

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1)2^k$$

Add  $(k+1)2^k$  term to b/s of  $\rightarrow$

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} + (k+1)2^k = 1 + (k-1)2^k + (k+1)2^k \\ = 1 + 2^k \{k-1 + k+1\}$$

$$= 1 + 2^k \cdot 2k$$

$$= 1 + k 2^{k+1}$$

which is of the form  $1 + (n-1)2^n$  when  $n$  is replaced by  $k+1$ .

$\therefore$  True for all  $n \geq 1$ .

4) a) i)  $\sqrt{3} \cos 2t - \sin 2t \equiv A \cos(2t + \alpha)$   
 $= A \cos \alpha \cos 2t - A \sin \alpha \sin 2t$   
 $\therefore \begin{cases} A \cos \alpha = \sqrt{3} \\ A \sin \alpha = 1 \end{cases} \Rightarrow \tan \alpha = \frac{1}{\sqrt{3}}$   
 $\alpha = \frac{\pi}{6}$

$A^2 = (\sqrt{3})^2 + 1^2 \Rightarrow A = 2$

$\therefore \sqrt{3} \cos 2t - \sin 2t \equiv 2 \cos(2t + \frac{\pi}{6})$

ii)  $2 \cos(2t + \frac{\pi}{6}) = 1$   
 $\cos(2t + \frac{\pi}{6}) = \frac{1}{2}$

$\therefore 2t + \frac{\pi}{6} = \pm \frac{\pi}{3} + 2n\pi$

$2t = \frac{\pi}{6} + 2n\pi \quad \text{or} \quad -\frac{\pi}{6} + 2n\pi$

$t = \frac{\pi}{12} + n\pi \quad \text{or} \quad n\pi - \frac{\pi}{12}$

b) i) Chord PQ:  $y = \frac{p+q}{2} x - apq$   
 Sub (v) a)  $a = 0 - apq$   
 $pq = -1$

ii) Tangent at P:  $y = px - ap^2$  — (1)

" " Q:  $y = qx - aq^2$  — (2)

(1) - (2)  $0 = (p-q)x - a(p^2 - q^2)$

$a(p-q)(p+q) = (p-q)x$

$p \neq q \therefore x = a(p+q)$

Sub into (1)

$y = ap(p+q) - ap^2$

$y = apq$

But  $pq = -1 \therefore y = -a$  as required.

iii) PQ is focal chord  $\therefore PQ = PS + SQ$

But  $PS = \text{dist of P to directrix} = ap^2 + a$

$SQ = \text{dist of Q to directrix} = aq^2 + a$

$\therefore PQ = ap^2 + a + aq^2 + a$   
 $= a(p^2 + 2 + q^2)$

But  $q = -\frac{1}{p} \therefore PQ = a(p^2 + 2 + \frac{1}{p^2})$   
 $= a(p + \frac{1}{p})^2$

4c) i)  $\angle APC = \angle ABC$  Angles in same segment standing on minor arc AC.

ii)  $\angle ABC = \angle ADE$  exterior angle of cyclic quadrilateral ADEB

$\therefore \angle APC = \angle ADE$  (Both equal to  $\angle ABC$ )

But  $\angle APC$  is exterior angle of quad ADKP & equals interior angle ADK.

$\therefore$  ADKP is a cyclic quadrilateral.

Q5 a)  $V = \pi \int_0^{\pi/4} \cos^2 x \, dx - \pi \int_0^{\pi/4} \sin^2 x \, dx$

$$V = \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx$$

$$V = \pi \int_0^{\pi/4} \cos 2x \, dx$$

$$= \pi \left[ \frac{\sin 2x}{2} \right]_0^{\pi/4}$$

$$= \frac{\pi}{2} (\sin \frac{\pi}{2} - \sin 0)$$

$$\underline{\underline{V = \frac{\pi}{2} \times 1}}$$

b)  $\ddot{x} = -16x$

i)  $\ddot{x} = \frac{d}{dx} \left( \frac{dx}{dt} \right) = -16x$

$$\frac{1}{2} v^2 = -\frac{16x^2}{2} + C$$

$$v^2 = -16x^2 + K$$

$x=1, v=4 \therefore 16 = -16 + K \Rightarrow K=32$

$$v^2 = 32 - 16x^2$$

$$= 16(2 - x^2)$$

$$v = \pm 4 \sqrt{2 - x^2}$$

$$\text{or } |v| = 4 \sqrt{2 - x^2}$$

ii) Max value of  $x$  for this to exist is  $x = \sqrt{2}$

5b) iii)  $\ddot{x} = -n^2 x$  with  $n=4$

$\therefore x = a \cos(4t + \alpha)$  is a general solution

$\therefore x = a \cos(4t + \alpha)$  But  $a = \sqrt{2}$

$x = \sqrt{2} \cos(4t + \alpha)$

$x = 1$  when  $t=0$

$1 = \sqrt{2} \cos \alpha$

$\frac{1}{\sqrt{2}} = \cos \alpha \Rightarrow \alpha = \pm \frac{\pi}{4}$

$x = \sqrt{2} \cos(4t \pm \frac{\pi}{4})$

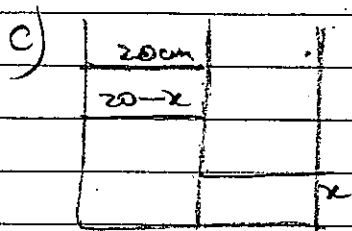
Now  $\dot{x} = -4\sqrt{2} \sin(4t \pm \frac{\pi}{4})$

$t=0, v=4 \therefore 4 = -4\sqrt{2} \sin(\pm \frac{\pi}{4})$

$\sin \pm \frac{\pi}{4} = -\frac{1}{\sqrt{2}}$

$\therefore$  Negative sign is required.

$\therefore x = \sqrt{2} \cos(4t - \frac{\pi}{4})$



When  $x$  is in one compartment  $20-x$  is in other. Difference =  $(20-x) - x = 20 - 2x$

$\therefore \frac{dx}{dt} = k(20 - 2x)$

ii)

$x = 10(1 - e^{-2kt}) = 10 - 10e^{-2kt}$

Differentiating w.r.t  $t$

$\frac{dx}{dt} = 10(0 - e^{-2kt} \cdot -2k)$

$= 10(2ke^{-2kt}) = 20ke^{-2kt}$

But  $x = 10 - 10e^{-2kt}$  or  $10e^{-2kt} = 10 - x$

$\therefore \frac{dx}{dt} = 2k(10 - x)$

$= k(20 - 2x)$  as required

iii)  $t=5, x=2 \therefore 2 = 10 - 10e^{-10k}$

$10e^{-10k} = 8$

$e^{-10k} = 0.8$

$-10k = \ln 0.8 \Rightarrow k = \frac{\ln 0.8}{-10} \approx 0.0223$

$$\text{Diff in levels} = 2 \text{ cm} \quad \therefore 20 - 2x = 2$$

$$18 = 2x \quad x = 9$$

$\therefore$  Sub  $x=9$  in

$$9 = 10(1 - e^{-2kt}) \quad x = 10(1 - e^{-2kt})$$

$$0.9 = 1 - e^{-2kt}$$

$$e^{-2kt} = 0.1$$

$$-2kt = \ln 0.1$$

$$t = \frac{\ln 0.1}{-2k}$$

$$t = \frac{\ln 0.1}{-0.0446} = 51.59 \text{ minutes.}$$

Q6 a)

$$\frac{dV}{dt} = 10 \text{ mm}^3/\text{s}$$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

$$10 = 500 \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{10}{500} = \frac{1}{50} \text{ mm/s}$$

b)  $v = \frac{32}{x} - \frac{x}{2}$

i)  $\ddot{x} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right) \quad \frac{1}{2} v^2 = \frac{1}{2} \left( \frac{32}{x} - 16 + \frac{x^2}{4} \right)$

$$\frac{1}{2} v^2 = 512 x^{-2} - 8 + \frac{x^2}{8}$$

$$\ddot{x} = -1024 x^{-3} + \frac{x}{4}$$

$$\ddot{x} = -\frac{1024}{x^3} + \frac{x}{4}$$

ii)  $\frac{dx}{dt} = \frac{64 - x^2}{2x}$

$$\frac{dt}{dx} = \frac{2x}{64 - x^2}$$

$$\therefore t = \int \frac{2x}{64-x^2} dx$$

$$t = -\ln(64-x^2) + c$$

$$t=0, x=2$$

$$0 = -\ln 60 + c$$

$$c = \ln 60$$

$$t = \ln 60 - \ln(64-x^2)$$

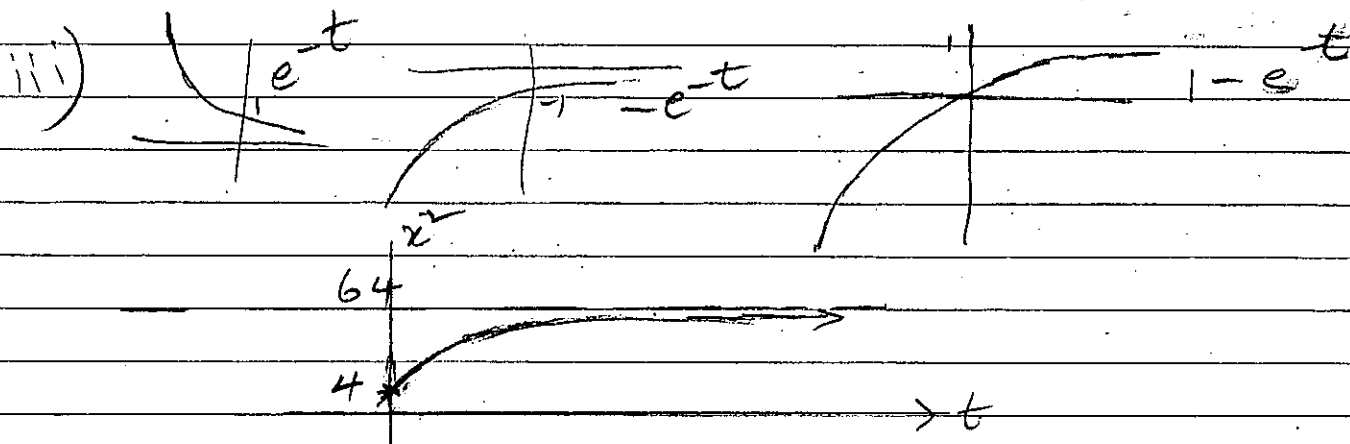
$$t = \ln \frac{60}{64-x^2}$$

$$\text{or } \frac{60}{64-x^2} = e^t$$

$$\text{or } \frac{64-x^2}{60} = e^{-t}$$

$$64-x^2 = 60e^{-t}$$

$$x^2 = 64 - 60e^{-t}$$



$$\text{as } t \rightarrow \infty \quad x^2 \rightarrow 64 \quad x \rightarrow 8$$

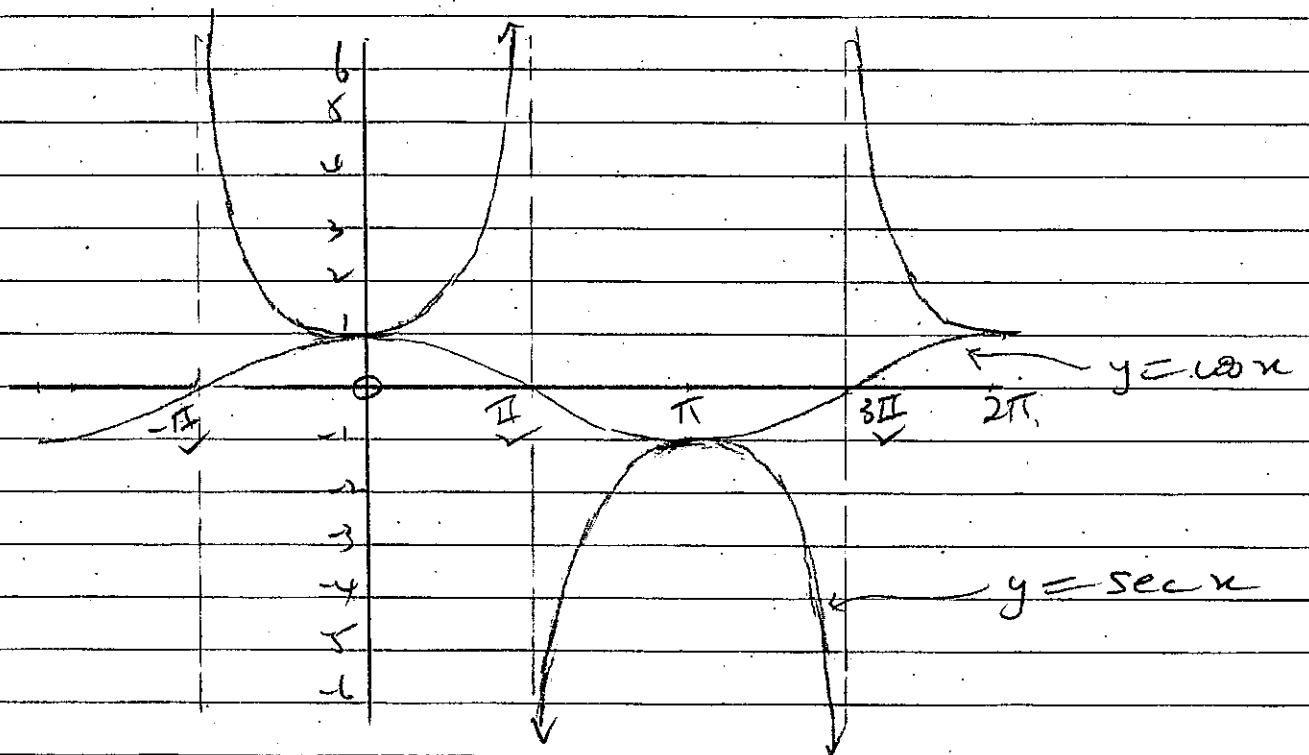
Note initial velocity  $v = \frac{32}{2} - \frac{2}{2} = 15 \text{ m/s}$  to right of 0.

$$\text{As } t \rightarrow \infty, x \rightarrow 8, v \rightarrow \frac{32}{8} - \frac{8}{2} \rightarrow 0$$

as  $x$  is always less than 8,  $v$  is positive  
 $\text{acc} \rightarrow \frac{-1024}{8^3} + \frac{8}{4} = -2 + 2 \rightarrow 0$  from negative. Particle is slowing down.

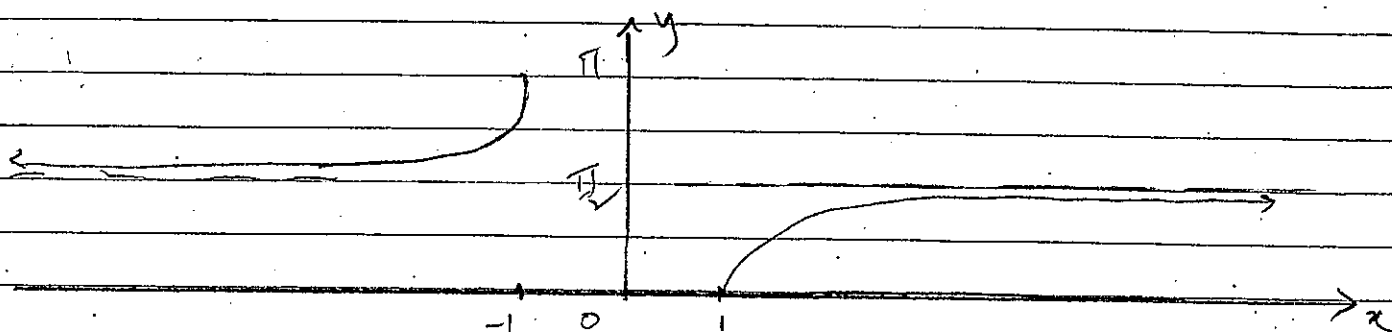
# Question 7

a)



Q3 b) i)  $\sec^{-1} \sqrt{2} = \theta \Rightarrow \sec \theta = \sqrt{2} \Rightarrow \theta = \frac{\pi}{4}$

ii)  $y = \sec^{-1} x$



DI  $x \geq 1$  or  $x \leq -1$

iii)

$x = \sec y$

$\frac{dx}{dy} = \sec y \tan y$

$\sec^2 y - \tan^2 y = 1$

$\frac{dy}{dx} = \frac{1}{\sec y \tan y}$

$\tan y = \sec y - 1$   
 $\tan y = \pm \sqrt{\sec y - 1}$

$\frac{dy}{dx} = \frac{1}{\sec y \cdot \pm \sqrt{\sec y - 1}}$

$\frac{dy}{dx} = \frac{\pm 1}{x \sqrt{x^2 - 1}} = \frac{1}{|x| \cdot \sqrt{x^2 - 1}}$

But Gradient is always positive,  $\therefore$  Absolute value sign is required when  $x < 0$  to give a positive value to  $\frac{dy}{dx}$

iv)  $y = \sec^{-1} \frac{x}{a}$

$\frac{dy}{dx} = \left| \frac{1}{\frac{x}{a} \sqrt{\frac{x^2}{a^2} - 1}} \right| \cdot \frac{1}{a}$

$= \left| \frac{a}{x \sqrt{\frac{x^2 - a^2}{a^2}}} \right| \cdot \frac{1}{a}$

$\frac{dy}{dx} = \left| \frac{a}{x \sqrt{x^2 - a^2}} \right|$

$$v) \int \frac{dx}{x\sqrt{25x^2-9}}$$

$$= \int \frac{dx}{5x\sqrt{x^2-\frac{9}{25}}}$$

$$\therefore a = \frac{3}{5}$$

$$= + \frac{1 \cdot \frac{5}{3} \sec^{-1} \frac{x}{3/5}}{5}$$

$$= \frac{1}{5} \cdot \frac{5}{3} \sec^{-1} \frac{5x}{3}$$

$$= \frac{1}{3} \sec^{-1} \frac{5x}{3} + C$$

(Once again positive sign required if  $x$  is positive, & negative when  $x$  is negative)